SOIL HEALTH NOW!



Science, technology and business intelligence for measuring, monitoring and increasing soil health

FULL AGENDA

7-10 December 2026

Center for the Dissemination of Research Results of the Aristotle University of Thessaloniki (KEDEA)

Thessaloniki, Greece

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AI 4 Soil Health

IBEC INTERBALKAN ENVIRONMENT CENTER

INVESTING IN REGENERATIVE AGRICULTURE AND FOOD

ARISTOTLE UNIVERSITY OF THESSALONIKI

Funded by the European Union

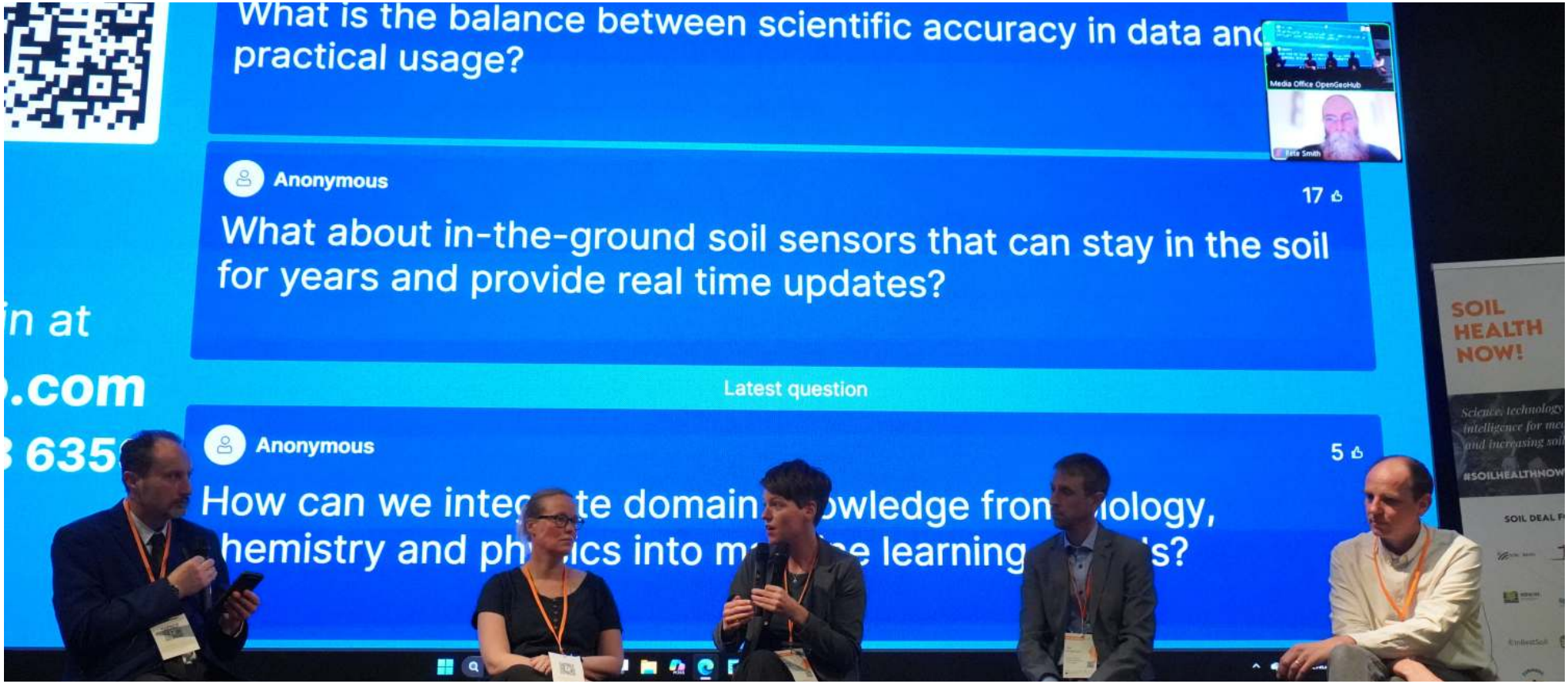
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OUR SOIL HEALTH NOW! 2025 WAS A SUCCESS

An immense thank you to all participants, presenters, keynotes and the soil health community for this first edition. It was an exciting experience having so many people whose perspectives could only make richer our original idea of having an urgent call for Soil Health in Europe. It proved a point, and we are proud to be soil stewards today and for future years. We hope we can bring once again in December a great time for you to delve deeper into multiple topics but with a common purpose: monitor, improve and generate long-lasting strategies to have healthy soils.

See you in December,

The Soil Health Now! team



TOPICS

This conference focuses on the urgent need for common soil health indicators, technology, and standards. This event aims to connect EU Agencies and Programs, researchers, farmer associations, technology leaders, and business intelligence to develop tools, protocols, standards, and datasets for measuring, monitoring, and increasing soil health across the European continent.

Specific topics of interest include:

- **Soil health indicators**
- **Pan-EU soil health assessment**
- **In-situ measurement of soil health**
- **Soil organic carbon**
- **Soil-climate-agriculture**
- **Carbon farming & regenerative agriculture**
- **Soil biology**
- **Technology for monitoring soil health**
- **Methods for reducing soil degradation**
- **Soil and nutrition nexus**
- **Soil Health beyond Europe**



ORGANIZING COMMITTEE

Beatriz Sarabia, *OpenGeoHub Foundation*
Gülten Inan, *OpenGeoHub Foundation*
Viola Coppi, *OpenGeoHub Foundation*
Tomislav Hengl, *OpenGeoHub Foundation*
Eva Maria Sarigiannidou, *i-BEC*
Dan Iles, *AI4SoilHealth Project*
Antonella Totaro, *RegenEarth*

DAY 1 - MONDAY, DECEMBER 7

	AMPHITEATHER I LIVE REC	AMPHITEATHER II LIVE REC	AMPHITEATHER III LIVE REC
09:00 - 10:00	*Registrations & morning coffee*		
10:00 - 10:30	Welcome speeches by Prof XXX. Vice Dean, (University of Thessaloniki); Tomislav Hengl (Opengeohub Foundation), Prof. George Zalidis (i-BEC)		
10:30 - 11.00	Keynote by Konstantinos Karantzalos (General Secretary Ministry of Digital Governance of Greece)		
11:00 - 11:30	Keynote: "Building Scalable Solutions for Soil Health" by George Zalidis (i-BEC, Aristotle University Thessaloniki)		
11:30 - 12:00	*Coffee Break*		
12:00 - 13:00	Keynote by Panos Panagos (EU Joint Research Centre)		
12:00 - 13:00	Keynote: "Understanding the Barriers to Regenerative Agriculture Adoption: A Behavioral Pathway Analysis for Greek Farmers" by Sheila Darmos (The Southern Lights)		
13:00 - 14:00	*Lunch Break*		
14:00 - 14:15	Oral Talk: "Deriving spatially explicit tolerable soil loss rates from a global synthesis of soil production rates" by Amy Thomas	Oral Talk: "From Field to Cloud: Operational AI Pipelines for Real-Time Soil-Health Measurements" by Theodoros Chalazas	Oral Talk: "Green Earth Program – Earth Observation-Supported Soil Rehabilitation in Drylands using Biological Soil Crusts" by Andreas Burr
14:15 - 14:30	Oral Talk: "Pan-European organic carbon trends in mineral agricultural soils – what do we know?" by Ali Sakhaee	Oral Talk: "Fusion of vis-NIR spectroscopy and ISE for online mapping of key soil fertility attributes" by Abdul Mounem Mouazen	Oral Talk: "Living Lab for Sustainable HOP Agroecosystem Health Monitoring Through Soil and Plant Microfungal Indicators" by Rasime Demirel
14:30 - 14:45	Oral Talk: "Using EO and remotely sensed data for irrigation management in the Cyprus Living Lab - The NEMESIS Horizon project" by George Papadavid	Oral Talk: "LIFE MONITOR: AN EMERGING METHOD FOR SOIL MONITORING AND QUALIFICATION" by Peter den Hartog	Oral Talk: "Development of a Multifunctional Biopreparation for Sustainable Maize Cultivation" by Justyna Szulc
14:45 - 15:00	Oral Talk: "High-Resolution Global Mapping of Peat Depth and Organic Soil Occurrence" by Mustafa Serkan Isik	Oral Talk: "Scaling Ecological Management Unit (EMU) Synthesis with a Domain-Specific AI Knowledge Engine" by Travis Nauman	Oral Talk: "Effects of Contaminated Soil on the Survival and Growth of European (Populus tremula L.) and Hybrid Aspen (Populus tremula L. × Populus tremuloides Michx.) Clones by Stand Density" by Mir Md Abdus Salam
15:00 - 15:15	Oral Talk: "Satellite-Based Assessment and Neural Network Prediction of Desertification Risk in Mediterranean Agricultural Landscapes" by Adrian Canovas-Rodriguez	Oral Talk: Oral talk: "SoilPATH: From evidence on arable farming practices to actionable soil-health decision support across Europe" by Emanuel Lekakis	Oral Talk: "Food processing residues as circular soil improvers for soil health" by Vera Proskynitopoulou
15:15 - 15:30	Oral Talk: "Soil Health Trends Index (SHTI): a pan-EU index quantifying soil health at 30 m resolution for 2000–2025 period" by Tom Hengl	Oral Talk: "The fraction of carbon in soil organic matter as a national-scale soil process indicator" by Sabine Reinsch	Oral Talk: "Monitoring Hidden Microfungal Biodiversity from Urban Forest to Agricultural Soils" by Rasime Demirel

Theme 1: Soil mapping from space & Pan-European monitoring

Theme 2: Theme 1: Soil mapping from space & Pan-European monitoring

Theme 3: Soil biology & restoration

DAY 1 - MONDAY, DECEMBER 7

	AMPHITEATHER I LIVE REC	AMPHITEATHER II LIVE REC	AMPHITEATHER III LIVE REC
15:30 - 16:00	*Coffee Break*		
16:00 - 16:15	Panel discussion 1: Food systems transformation in South-eastern Europe and the Balkans (i-BEC)	Workshop: "From eDNA to Decisions: A Live Demonstration of the Soilytix MySoil Dashboard for Soil Biodiversity Monitoring and MRV" by Julien Winkel	
16:15 - 16:30			
16:30 - 16:45			
16:45 - 17:00			
17:00 - 20:00	Poster and networking session + Drinks and Bites		

DAY 2 – TUESDAY, DECEMBER 8

	AMPHITEATHER I (LIVE REC)	AMPHITEATHER II (LIVE REC)	AMPHITEATHER III (LIVE REC)
08:00 - 09:00	*Registrations & morning coffee*		
09:15 - 09:30	Recap & run of the day		
09:30 - 10:00	Keynote: "Bridging Soil Data and Real farm Decisions?" by John Gilliland (Arc Zero)		
10:00 - 10:30	Keynote: "Pan-European Soil Health Indicators: Progress, Challenges, and Opportunities from the AI4SoilHealth Project" by David Robinson (UKCEH)		
10:30 - 11.00	Keynote: "Carbon Farming Hub: Bringing Carbon Farming into Practice Through Regenerative Agriculture" by Pietro Baldrighi (Confagricultura ANGA)		
11:00 - 11:30	*Coffee Break*		
11:30 - 12:00	Keynote: "From Field to Lab: Integrating In-Situ and Laboratory Indicators for Soil Health Assessment" by Fatemeh Hateffard (Stockholm University)		
12:00 - 13:00	Panel discussion 2: Carbon Farming at scale: How to bring real land managers' needs and research together (RegenEarth x OGCR project)		
13:00 - 14:00	*Lunch break*		
14:00 - 14:15	Oral Talk: "In Situ Soil Health Measurement to Support Regenerative Grazing Practices Across European Living Labs (GroundWork project)" by Justine Luca	Oral Talk: "Building African Union Soil Observatory" by Thaïsa van der Woude	Oral Talk: "A Digital Twin of Ecosystem for the climate-biodiversity nexus: the TerraTwin project. Soil biodiversity and compound flooding, the Belgian Use Case" by Radu Giurgiu, Theodoros Chalazas
14:15 - 14:30	Oral Talk: "A comparison of model performances and uncertainty of soil remote sensing products calibration for large-scale carbon farming accounting" by Maria Fantappie, Carlos Lozano Fondón	Oral Talk: "SHERPA in Action: Soil Health Quantification on European scale across different land cover and use" by Christine Alewell	Oral Talk: "Soil Community Attributes as Indicators of Soil Health under Anthropogenic and Environmental Pressures" by Nikolaos Monokrousos
14:30 - 14:45	Oral Talk: "Sample Size Requirements for Carbon Farming Projects with Baseline Control Sites" by Erik Scharwächter	Oral Talk: "Shaping NextSoils+ Soil Health Services with Stakeholders for the Sentinel Expansion Missions" by Konstantinos Karyotis	Oral Talk: "Organic carbon dominates soil porosity across environmental gradients, with OC–clay balance providing secondary structural insight" by Amy Thomas
14:45 - 15:00	Oral Talk: "Agricultural liming: an overlooked climate-smart practice for soil health and carbon farming in Europe" by Ana Dutton	Oral Talk: "Implications of total and active soil microbiome dynamics for long-term soil health monitoring" by Lur Epelde	Oral Talk: "From Funding Individual Projects to Building a Territorial Innovation Portfolio: Lessons from 25 Experimental Sites in the French Landes Living Lab" by Wafa Guiga
15:00 - 15:15	Oral Talk: ""Multi-Faceted Biodiversity Integration for a Healthy Soil and Resilient Crop Management - SOILRES" by Davide Cammarano	Oral Talk: "Advancing soil monitoring through citizen science-generated data" by Céline Laurent	Oral Talk: "A framework for soil health assessment and monitoring across scales" by Lucas Gomes
15:15 - 15:30	Oral Talk: ""Toward a Digital Twin for Agriculture: Soil Health and Carbon Assessment with MODASYS" by Mohammad Ibrahim Khalil	Oral Talk: " "SoilHive: An Open-Source Platform for Global Soil Data Harmonisation" by Ester Miglio	Oral Talk: "Cluster-Informed Soil Spectroscopy for Predicting Soil Health Indicators Across the European Pilot Sites of AI4SoilHealth" by Konstantinos Karyotis

Theme 1: Agricultural monitoring & Carbon Farming

Theme 2: Global Soil Data & Digital Infrastructure

Theme 3: Soil biodiversity, SOC & Climate

DAY 2 - TUESDAY, DECEMBER 8

	AMPHITEATHER I LIVE REC	AMPHITEATHER II LIVE REC	AMPHITEATHER III LIVE REC
15:30 - 16:00	*Coffee Break*		
16:00 - 16:15	SOB4ES special session: 1. Oral talk: "Linking Soil Biodiversity, Environmental Constraints and Ecosystem Services across European Landscapes using Earth Observation and Machine Learning" by Maria Marily Christou 2. Oral talk: "Data-driven ecosystem management: assessing soil carbon responses to land use and restoration practices using spectroscopy and digital soil mapping" by Karen Vancampenhout, Sam Ottoy	Workshop: "Make your soil data or knowledge record findable: a hands-on SoilWise publishing workshop" by Radu Giurgiu, Wannes De Man	Workshop: "Measure soil enzymatic activity (right here in the room) and learn why you should do it more" by Sonia Meller
16:15 - 16:30			
16:30 - 16:45			
16:45 - 17:00			
19:00 - 21:00	Social Dinner		

DAY 3 – WEDNESDAY, DECEMBER 9

Day theme: Living labs, Future Soil Health Monitoring & Earth Observation

	AMPHITEATHER I LIVE REC	AMPHITEATHER II LIVE REC	AMPHITEATHER III LIVE REC
08:00 - 09:00	Registrations & *morning coffee*		
09:15 - 09:30	Recap & run of the day		
09:30 - 10:00	**Keynote:** "Why human health starts in the soil? by Eirini Tsirimokou (Regenerative Healthcare European Association)		
10:00 - 10:30	**Keynote:** "On the Value of Soil Knowledge in the Era of Data-Driven Soil Modelling by Sarem Norouzi (Aarhus University)		
10:30 - 11.00	**Keynote:** "African Union Soil Observatory (AUSO): Innovative delivery of AI-ready datato power continental soil health policy implementation and soil intelligence systems" by Christian Omuto (FARA)		
11:00 - 11:30	*Coffee Break*		
11:30 - 12:00	**Keynote:** "Setting up the Danish Monitoring Infrastructure" by Mogens Humlekrog Greve (Aarhus University/AI4SoilHealth)		
12:00 - 13:00	**Panel discussion 3:** Soil Monitoring Law (SML) Implementation Challenges (AI4SoilHealth project)		
13:00 - 14:00	*Lunch break*		
14:00 - 14:15	**Oral Talk:** "SoilWise - Advancing a fair data and knowledge infrastructure for healthier soils in Europe" by Radu Giurgiu	**Oral Talk:** "Scores of Visual Evaluation of Soil Structure (VESS) as a Soil Health Indicator Across Contrasting Land Use Types and Soil Textures" by Niklas Schmücker	**AI4SOILHEALTH PROJECT FINAL MEETING** & SPECIAL SESSION
14:15 - 14:30	**Oral Talk:** "The Role of the Western Macedonia Living Lab in Soil Health: From Co-Creation to Harmonized Monitoring" by Savvas Karagiorgos	**Oral Talk:** "A comparison of DSM-based SOC stock estimators: Uncertainty reductions and practical insights" by Erik Scharwächter, Ahmad Awad	
14:30 - 14:45	**Oral Talk:** "Patterns and thresholds for soil pH across Europe in relation to soil health and degradation" by Inma Lebron Robinson	**Oral Talk:** "Linking soil organic matter, bulk density and runoff thresholds under extreme rainfall: a modelling study using GB national soils data" by Maud van Soest	
14:45 - 15:00	**Oral Talk:** "Influence of Biochar, Limestone, and Their Combination on Cadmium and Zinc Immobilization in Metal-Contaminated Boreal Soils" by Irini Papageorgiou	**Oral Talk:** "Towards High-Resolution Soil Type Mapping across Europe Using Machine Learning and Earth Observation Data" by Tomislav Hengl	
15:00 - 15:15	**Oral Talk:** "Living Labs as engines for soil health governance, monitoring, harmonization, and the road ahead" by Ioannis Varvaris, Diofantos Hadjimitsis, Georgeo Papadavid	**Oral Talk:** "SUS-SOIL: Project Objectives, Implementation Progress and Perspectives for Sustainable Soil Management" by Anastasia Pantera	
15:15 - 15:30	**Oral Talk:** "From Soil Monitoring to Soil Intelligence: Building Governance-Ready AI Systems for Tropical Agricultural Landscapes" by Suzie Haryanti Husain	**Oral Talk:** "Decadal soil organic carbon and nitrogen decline across contrasting arable management in a long-term experiment in Northern France" by Aude Mancia	

DAY 3 - WEDNESDAY, DECEMBER 9

	AMPHITEATHER I LIVE REC	AMPHITEATHER II REC	AMPHITEATHER III REC
15:30 - 16:00	*Coffee Break (Break Area, Basement floor)*		
16:00 - 16:15	**Workshop 5:** Help shape the African Union Soil Observatory (AUSO): share your user perspective and expert feedback by Eng. Thaïsa van der Woude (ISRIC), Prof. Christian Omuto (FARA), Dr. David Robinson (UK-CEH), Dr. Caity Tanner (CABI)	**Workshop 5:** "From Soil Health Indicators to Action: What Can We Learn from Europe's Soil Living Labs?" by Marc Rosiers (European Landowners' Organization - ELO)	**AI4SOILHEALTH PROJECT FINAL MEETING** & SPECIAL SESSION
16:15 - 16:30			
16:30 - 16:45			
16:45 - 17:00			
17:00 - 17:15	**End of the program**		

FIELD EXCURSION TO KTIMA GEROVASSILIOU VINEYARD – AI4SOILHEALTH PILOT SITE

The field excursion to **Domaine Gerovassiliou**, located in the wine-growing region of Epanomi, approximately **40–60 minutes from Thessaloniki,** offers participants the opportunity to experience how soil health monitoring and regenerative viticulture are applied in practice. Internationally recognised for its commitment to quality winemaking and environmental stewardship, the estate is also home to one of the **AI4SoilHealth pilot sites**, where innovative soil monitoring methods and sustainable vineyard management practices are being demonstrated.

During a guided tour of the vineyard, participants will explore how soil health indicators and digital technologies support decision-making, while learning about the regenerative practices implemented to enhance soil functions, biodiversity, and long-term resilience.

What to expect during the excursion:

AI4SoilHealth Pilot Site: Discover how soil health is monitored using innovative methods and how the resulting data support sustainable vineyard management.
Regenerative Viticulture: Learn about the practices implemented to improve soil health, biodiversity, and ecosystem resilience in a working commercial vineyard.
Guided Vineyard Tour: Explore the estate with local experts and gain insight into its history, viticultural practices, and the relationship between healthy soils and wine production.
Wine Tasting: Conclude the visit with a guided tasting of a selection of Domaine Gerovassiliou wines, showcasing the distinctive character of the region.

Transportation will be provided from Thessaloniki, with the excursion lasting approximately 4 hours, including travel, the vineyard visit, and the wine tasting, before returning to the city in the early afternoon.



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POSTER SESSIONS

Soil health monitoring in nature conservation across European protected areas, Jasmin Liedtke

SOB4C: The relative importance of biotic and abiotic drivers of soil carbon stocks at the European scale, Astrid Sneyders

Assessing Soil Health Across Europe Using Collembola Diversity Indicators, Fiera Cristina

Use and Substitution Potential of Agricultural Plastics in Apple Orchards, Mirjam Hofer

LandPrint: Mapping the Human Imprint on Soil Health Drivers, Costs, and Solutions Across Europe, Georgios Chatzichristos, Emanuel Lekakis, Sofia Magopoulou, Eleni Kalopesa

Selecting Phytoremediation Strategies for Acidic Boreal Forest Soils: A Comparison of Chelator-Assisted Phytoextraction and Biochar-Assisted Phytostabilization Across Contamination Levels, Darshani Kumaragamage Srimathie Indraratne

SOC enrichment and GHG mitigation in Greek Mediterranean vineyard, Paraskevi Chantzi, Daphne Kitsou

Multi-Species Cover Crops in Regenerative Agriculture Enhance Soil Quality and Weed Suppression in Mediterranean Cropping Systems, Christos Vasilikiotis

Reassessing Cobalt Fertilization in Agricultural Soils: Implications for Soil Health and Pollution Prevention, Efie Tsourgianni

Soil fauna responses to agricultural management across Europe: a meta-analysis of biological metrics and decomposer taxa, Nga Thi Thanh Mai

Building the MRV4SOC Knowledge Hub: An Interoperable Digital Platform for Soil Monitoring, Data Integration, and Open Science, Dimitra Palantza

UAV-Based Hyperspectral SOC Mapping Using a Tunable Fabry-Pérot Imager and Machine Learning, Wannes De Man, Haris Ampas, Tuna Coppens, Nick Berkvens

SoilWise project: the journey of a FAIR metadata record from publication to reusability, Radu Giurgiu, Wannes De Man

Cultivating Soil Literacy: Bridging the Science-Society Gap through Arts and Participatory Engagement, Mackenzie Baert

Towards a Soil Resilience and Food Security Index for Climate-Resilient Agricultural Landscapes, Konstantinos Karyotis, Maria Marily Christou

Setting up Agro Living Labs: Focus on the Financial Support to Third Parties (FSTP) process to support on-the-ground research, Clémentine Eynard

Adapting NIR Spectroscopy for Farm-Scale Soil Organic Carbon Monitoring, Chris Collins, Jessica Underwood

Edge-AI Autoencoder Anomaly Detection for Trustworthy Electrochemical Sensor Signals in Soil-Health Monitoring, Aristeidis Papadopoulos

Soil Degradation Assessment in Europe: Historical Trends and Future Projections by Mehdi H. Afshar

PARTNERSHIPS WITH HORIZON EUROPE AND OTHER SOIL HEALTH PROJECTS

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